YAKEEN

NEET 2.0 2027

Physics | Lecture - 11

Motion in a Straight Line

By- Manish Raj (MR Sir)



Todays Goal

↳ Special graph & graph conversion.

H/w goal

→ 1-D Sangh Assign

1 to 5 → *

→ Revision → Vector upto 4-lecture

Question

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37

A particle starts from rest with constant acceleration. The ratio of average velocity to the time average velocity is: [2000]

1. 1/2
2. 3/4
3. 4/3
4. 3/2

$u=0$ $a=$ const.

$v^2=u^2+2ax$

Put $u=0$ $\Rightarrow v=\sqrt{2ax}$

$$\langle v\rangle_{\text{space Avg}}=\frac{\int v\,dx}{\int dx}$$

$$=\frac{\int\sqrt{2ax}\,dx}{\int dx}$$

$$=\frac{\sqrt{2a}\int x^{1/2}\,dx}{x}=\frac{\sqrt{2a}\,x^{3/2}x^{1/2}}{\frac{3}{2}x}=\frac{2\sqrt{2ax}}{3}=\frac{2v}{3}$$

$v=u+at$ ($u=0$)

$$\langle v\rangle_{\text{time Avg}}=\frac{\int v\,dt}{\int dt}=\frac{\int at\,dt}{\int dt}$$

$$=\frac{a\,t^2t}{2t}=\frac{at}{2}=\frac{v}{2}$$

$$\frac{\langle v\rangle_{\text{space}}}{\langle v\rangle_{\text{tim}}}=\frac{2v}{3\times\frac{v}{2}}=\frac{4}{3}$$

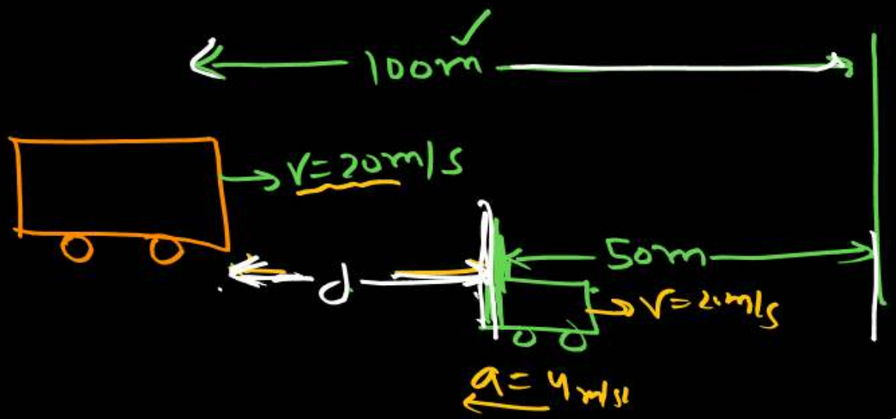
Question

Copy in notz

(38)

A car is moving with 20 m/s suddenly driver of car observes a stationary truck ahead of 100 m distance after some reaction time drive apply a break of conrt. retardation 4 m/s^2. find out maximum reaction time to prevent the collision? =?

Sol^n



$S_{(stopping\ distance)} = \frac{u^2}{2a} = \frac{10 \times 10}{2 \times 4} = 50m$

d=50m

a=0

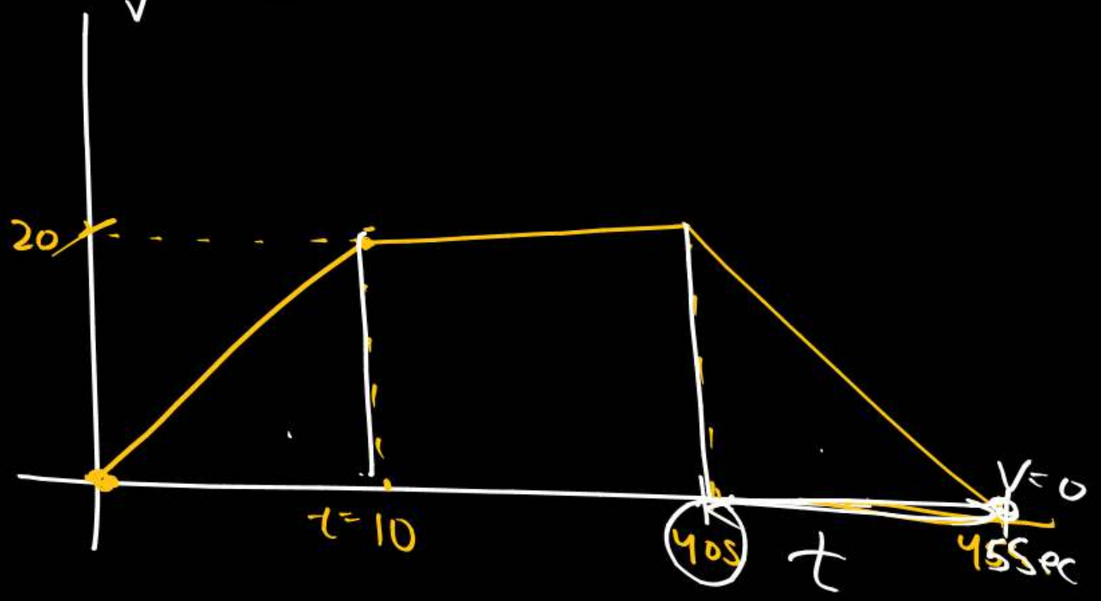
$t = \frac{d}{u} = \frac{50}{20}$

$= 2.5\ sec$

Question

A particle starts from rest, accelerates at 2 m/s^2 for 10s and then goes with constant speed for 30s and then decelerates at 4 m/s^2 till it stops. What is the distance travelled by it

1. 750 m
2. 800 m
3. 700 m
4. 850 m



Area = $\frac{1}{2}[30+45) \times 20$

$= 75 \times 10$

$= 750m$

$u=0$
$a=2$
$t=10sec$
$V=u+at = 0+2\times10 = 20$

$V=0$
$u=20$
$a=-4m/s^2$
$t=5sec$

$V=u+at$
$0=20-4t$
$t=5s$

Question

The engine of a motor cycle can produce maximum acceleration 5 m/s^2. It can produce maximum retardation 10 m/s^2 what is then minimum time in which he will cover a distance of 1.5 km.

$a = 5\,m/s^2$ ✓

$\beta = -10\,m/s^2$ ✓

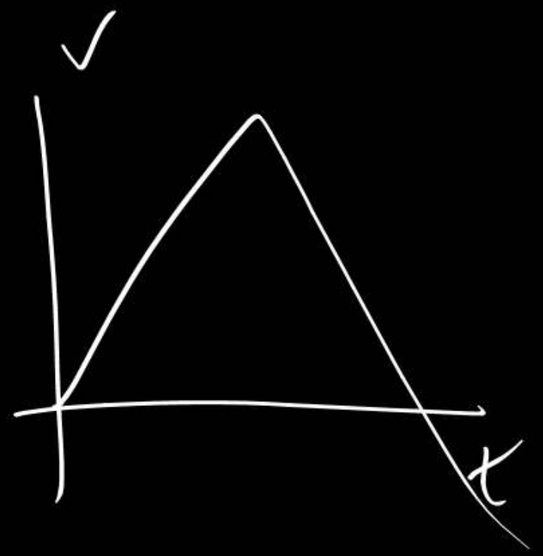
1. 5 sec
2. 15 sec
3. 10 sec
4. 30 sec ✓

$$S = \frac{1}{2}\frac{\alpha\beta}{\alpha+\beta}T^2$$

$$1500\,m = \frac{1}{2}\frac{5\times10}{(15)}T^2$$

$$60\times15 = T^2$$

$$T = \sqrt{900} = 30\,s$$



Question

A car start from rest and moves with acceleration = 10 m/s^2 for time 5 sec after this it retards with $-5\ m/s^2$ and comes to at rest then find total distance.

Sam-

$$\alpha t_1 = \beta t_2$$

$$S = \frac{1}{2}\frac{\alpha\beta}{\alpha+\beta}t^2$$

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object moving with 20m/s constⁿ velocity for 5-sec after that it retards with $-10m/s^2$ for 4-sec then moves with constant velocity for 2-more sec find distⁿ & dispⁿ

Question

M/W Kal discuss

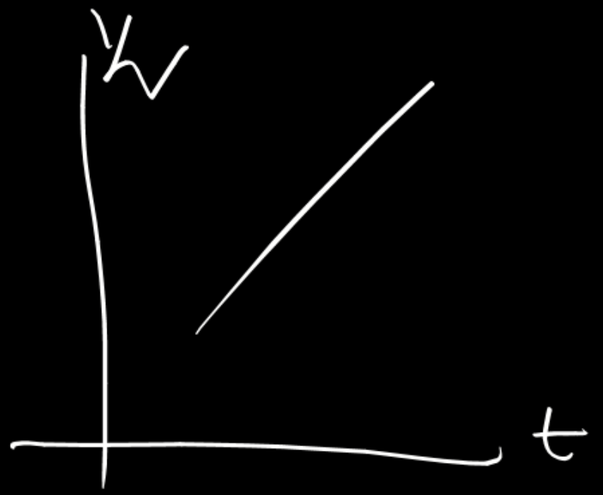
A car starts from rest and moves with uniform acceleration a on a straight road from time $t = 0$ to $t = T$. After that, a constant deceleration brings it to rest. In this process the average speed of the car is

1. aT
2. $\frac{aT}{4}$
3. $\frac{3aT}{2}$
4. $\frac{aT}{2}$

$= v\frac{dv}{dx}$

Area ✗ | $x \xleftarrow{\text{Am}}$ slope $\rightarrow v \xleftarrow{\text{Area}}$ slope $\rightarrow$ slope लगाओ

	Graph	slope	Area	y-component	x-component
❤ 1.	$x-t$ Position – time	slope $= \frac{dx}{dt} = v$	✗	Position	time
❤ 2.	velocity – time $v-t$	slope $= \frac{dv}{dt} = a$	Area $= \int v\,dt$ = disp	velocity	time
3. *	velocity – Position $v-x$	slope $= \frac{dv}{dx} = \left(\frac{a}{v}\right)$	✗	velocity	Position
4. ✓ *	$v^2 - x$	slope $= \left(\frac{dv^2}{dx}\right) = 2v\frac{dv}{dx} = 2a$ acc^n	✗	v^2	"
❤ 5.	acc^n – time $a-t$	✗	Area $= \int a\,dt$ = change in vel $= v-u$	acc^n	time.
6. *	acc^n Position $a-x$	✗	Area $= \int a\,dx = \int v\frac{dv}{dx}dx = \int_u^v v\,dv = \left(\frac{v^2}{2}\right)_u^v = \frac{v^2-u^2}{2}$		Positi.



(41) Slope of this graph will be ??

Soln

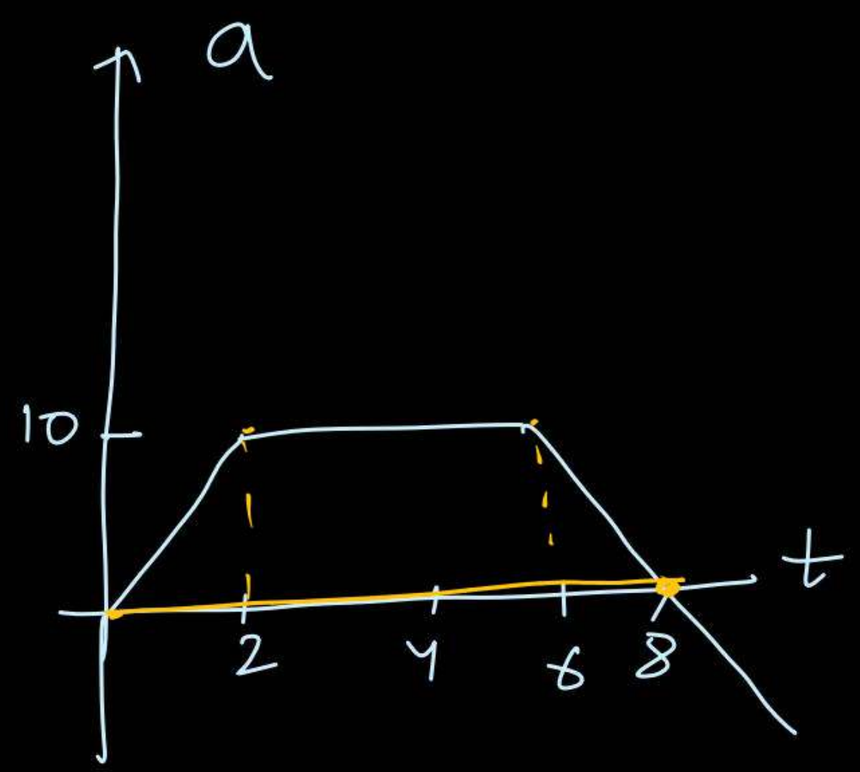
$$\text{Slope} = \frac{dv^{-1}}{dt}$$

$$= \frac{dv^{-1}}{dv} \times \frac{dv}{dt}$$

$$\text{Slope} = -1(v^{-2})\,a$$

$$(\text{Slope}) = -\frac{a}{v^2}$$

(42) If intial Velocity of object $-10 m/s$ then find velocity at $t = 8 sec$



Sol^{n}

$$\text{Area} = \vec{v} - \vec{u}$$

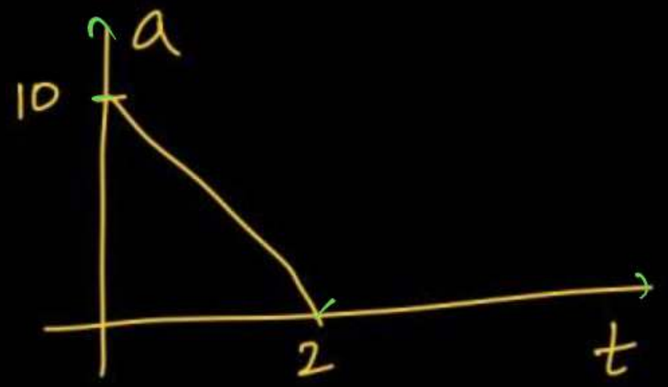
$$\frac{1}{2}[8+4) \times \frac{5}{1} = \vec{v} - (-10)$$

$$\Rightarrow \quad 60 = v + 10$$

$$v = 60 - 10 = 50 m/s$$

(12)

acceleration - time graph:-



if intial velocity is 2 m/s then find velocity at $t = 2$ sec

$V - u$ = Area

$V - 2 = \frac{1}{2} \times 10 \times 2$

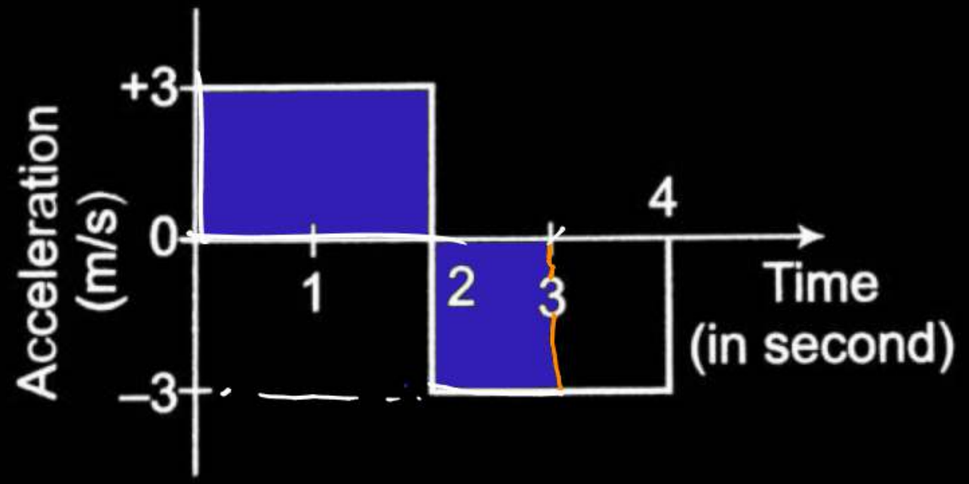
$V = 10 + 2 = 12$ ✓

⇒ Area of a/t graph is velocity



Question

A particle starts from rest at $t = 0$ and moves in a straight line with an acceleration as shown below. The velocity of the particle at $t = 3s$ is



1. 2 m/s
2. 3 m/s
3. 4 m/s
4. 6 m/s

$v - u$ = Area

$v = u + Am$

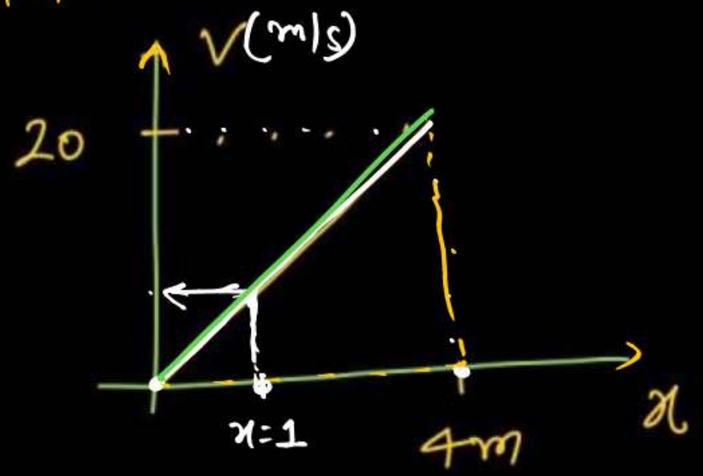
$= 0 + 3 \times 2 - 3 \times 1$

$= 6 - 3 = 3$ m.

(43)

find accn at $x = 1m$

copy / mark sir



$$\text{Slope} = \frac{20}{4} = 5$$

" "

$$y = mx + c$$

$$V = 5x + 0$$

$$V = 5 \times (1)$$

$$V = 5 \quad (x = 1)$$

$$\text{Slope} = \frac{dv}{dx} = \frac{a}{V}$$

$$\text{slope} = \frac{a}{v}$$

$$a = [\text{slope} \times V]$$

$$a = 5 \times 5$$

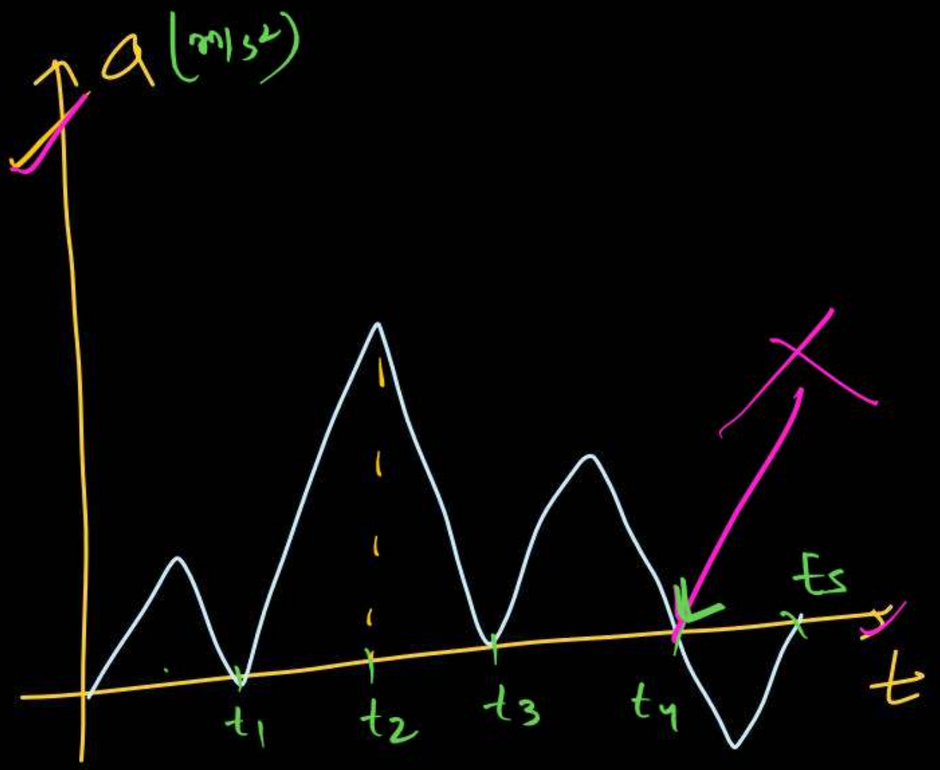
$x = 1m$

$$= 25 m/s^2$$

hence accn is variable

a = slope $(v) \uparrow$

call



(44) acc^n is max^m at.

(a) t_1

(b) t_2 (72%)

(c) t_3

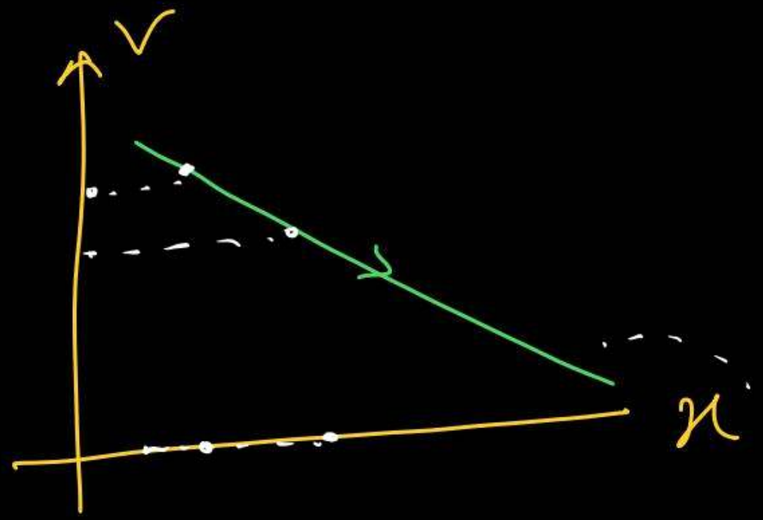
(d) t_4

(e) t_5

(45) In above question velocity is max^m at = ??

(d) t_4 Ans

(Q6) Comment on variation of magnitude of acceleration



(a) Const^n

(b) Increasing

(c) decreasing

Soln

$$\text{Slope} = \frac{dv}{dx}$$

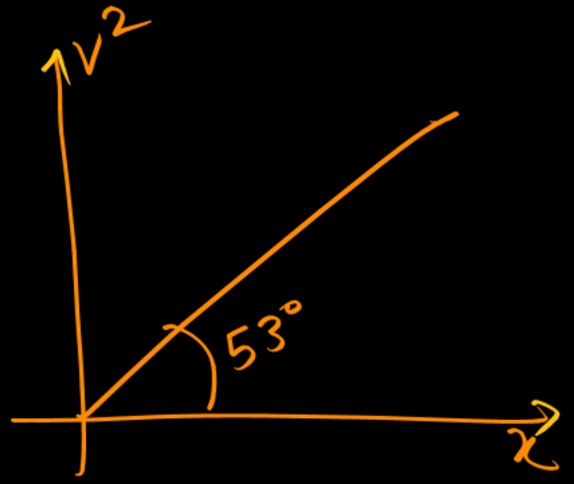
$$\text{Slope} = \frac{a}{v}$$

$$a = v(\text{slope})$$

$$a = \downarrow(-ve\ \text{slope})$$

47

Copy



find accn of the object in given graph ??

Sol^n

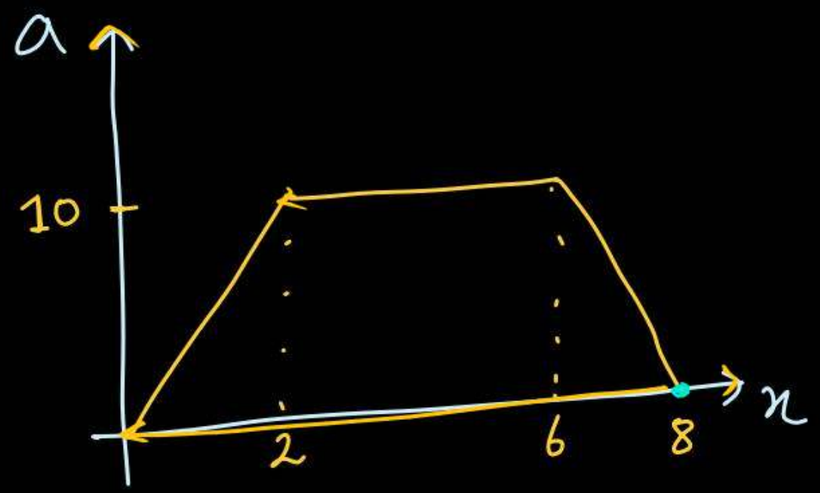
$$\text{Slope} = \frac{dv^2}{dx}$$

$$\tan 53^\circ = 2v\frac{dv}{dx}$$

$$\frac{4}{3} = 2a$$

$$a = \frac{2}{3}\ m/s^2$$

(17) If intial velocity is 1 m/s then find velocity at $x = 8m$



$u = 1\ m/s$

$V_{x=8m} = ??$

Soln

$$\text{Area} = \frac{v^2 - u^2}{2}$$

$$\frac{5}{10} \times \frac{1}{2}[8+4] = \frac{v^2 - (1)^2}{2}$$

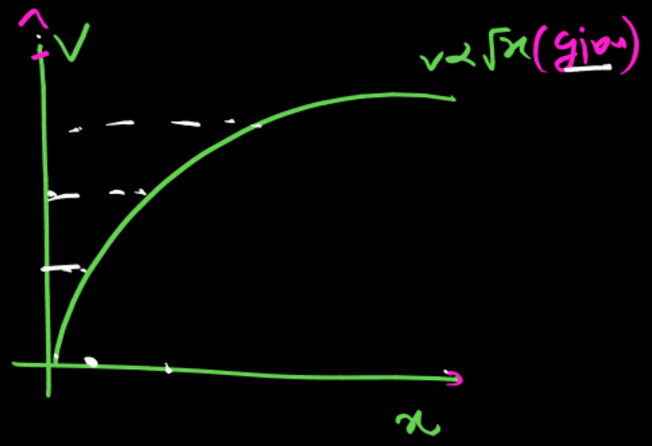
$$60 = \frac{v^2 - 1}{2}$$

$$120 = v^2 - 1$$

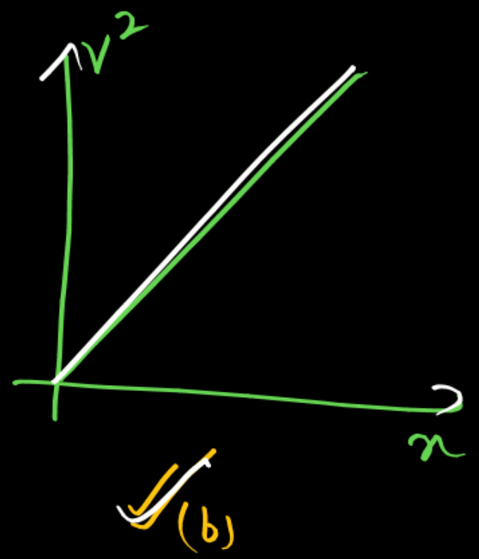
$$\# \quad V = \sqrt{121} = 11\ m/s$$

+1 (49)

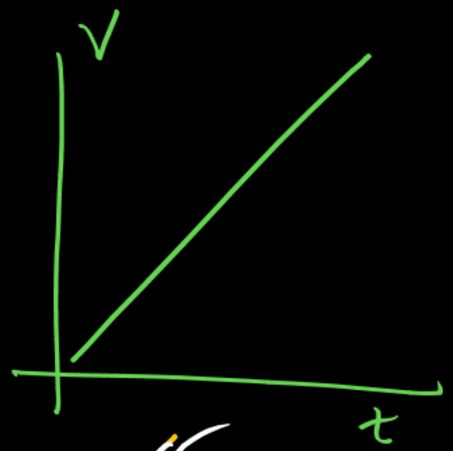
In which graph acc^n of object is const^n.



(a)



(b)



(c)

(d) all of these (Not)

Slope = acc^n ✓

Slope = 2 × acc^n

$\text{acc}^n = \left(\frac{\text{slope}}{2}\right) = a$

Method-1

$\# V = k\sqrt{x}$

$a = v\frac{dv}{dx}$

$= k\sqrt{x}\,\frac{d(k\sqrt{x})}{dx} = k^2\sqrt{x}\,\frac{1}{2\sqrt{x}}$

$= \left(\frac{k^2}{2}\right) = \text{cn.}$

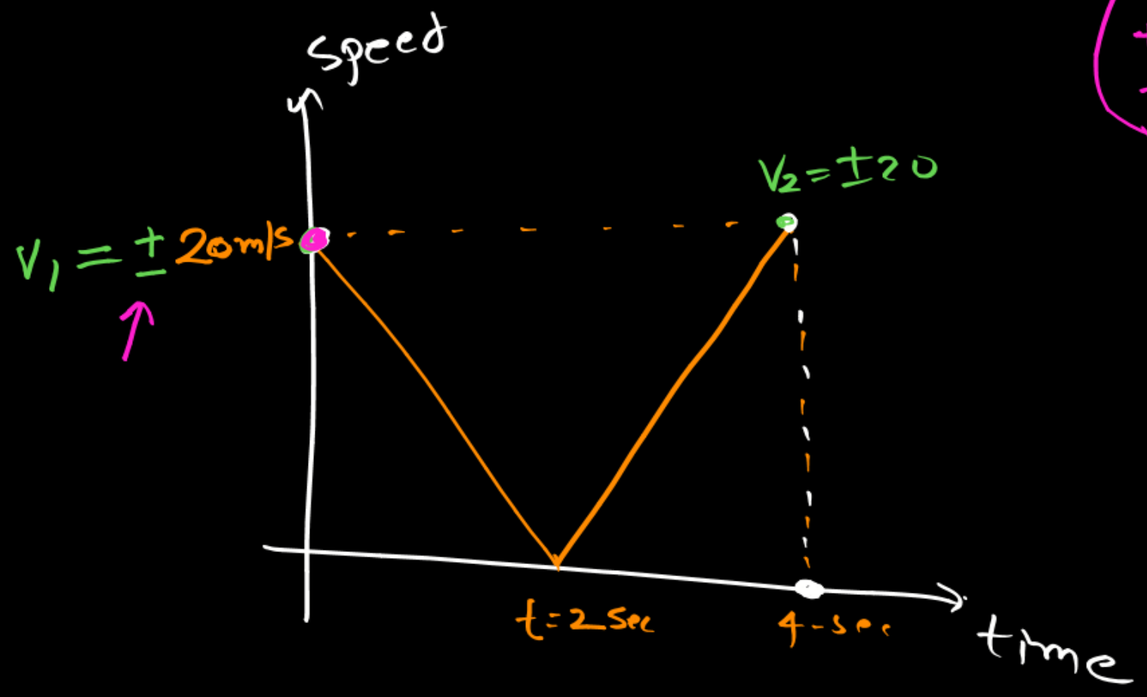
Method-2

$\text{Slope} = \frac{dv}{dx} = \frac{a}{v}$

$a = \uparrow v(\text{slope})\downarrow = \text{const}^n$

MRQ (50)

speed / time graph is given then find value of Average accⁿ in 4-sec



v_i	v_f
+20	+20
-20	-20

(a) $5 m/s^2$ ✗

(b) $10 m/s^2$ ✗

(c) zero ✗

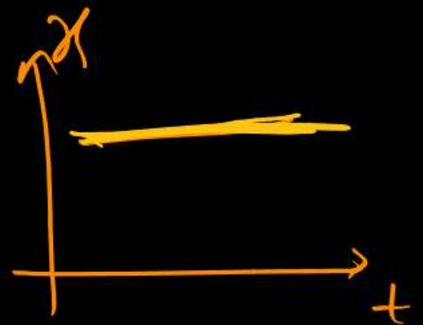
(d) $-10 m/s^2$ ✗

(e) b, c & d are correct Ans (+1)

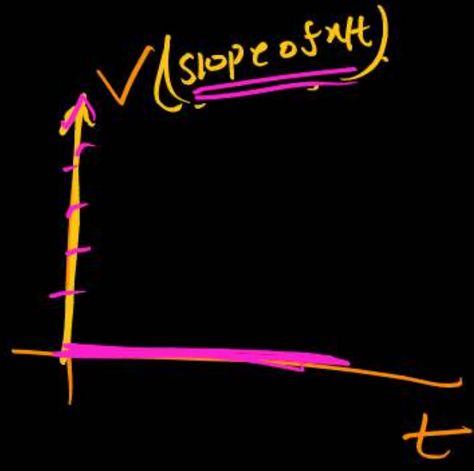
$$\text{Avg acc}^n = \frac{\vec{V}_f - \vec{V}_i}{\Delta t} = 0$$

$$= 0 \quad \frac{20-(-20)}{4} = +10 \quad \frac{-20-(20)}{4} = -10 m/s^2$$

(51)

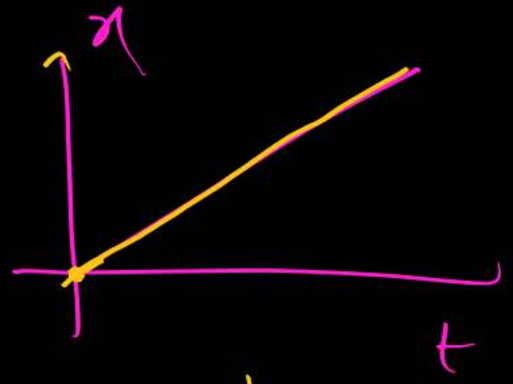


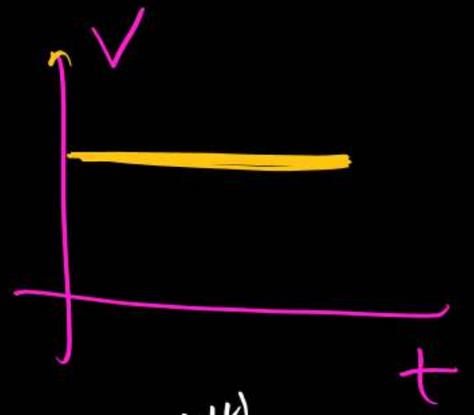
Convert into v/t



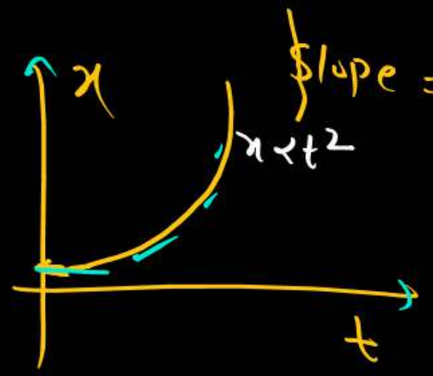
slope of x/t = v

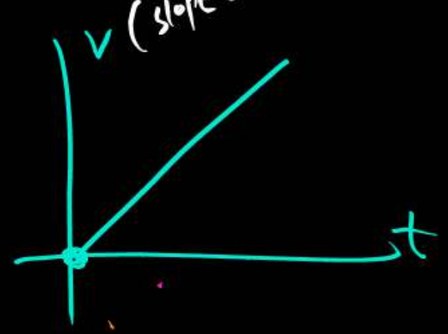
(52)



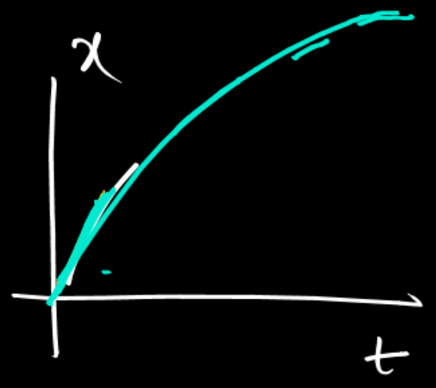


(53)

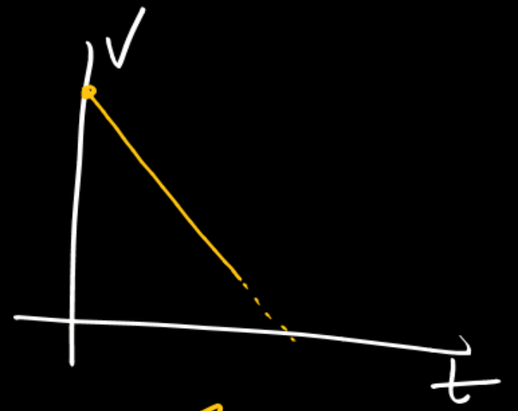




(54)



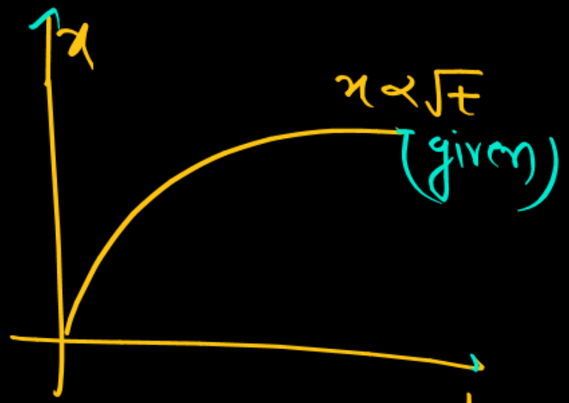
Convert into v/t



Slope → +ve
& decreasing

If equation of x/t is not known then this is correct ↗ then best possible

(55)

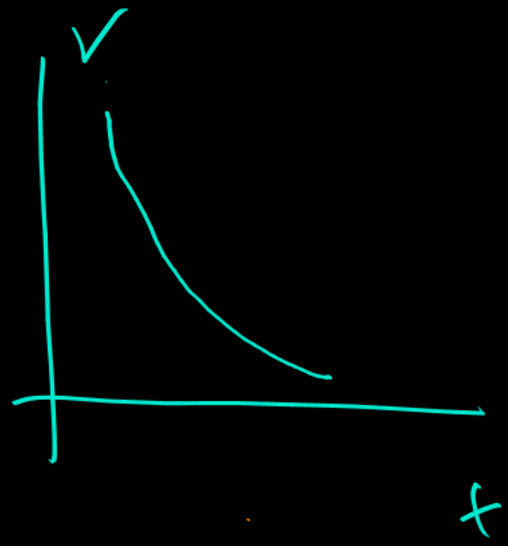


$$V = \frac{dx}{dt} = \frac{d\sqrt{t}}{dt}$$

$$= \frac{1}{2} t^{1/2-1}$$

$$= \frac{1}{2} t^{-1/2}$$

$$V = \frac{1}{2\sqrt{t}}$$



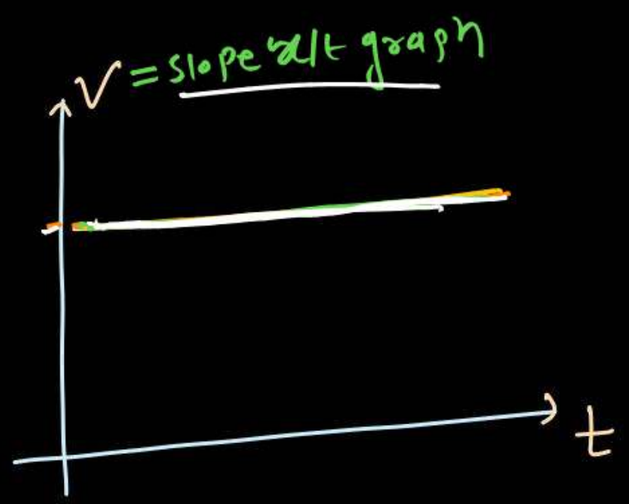
Graph Conversion

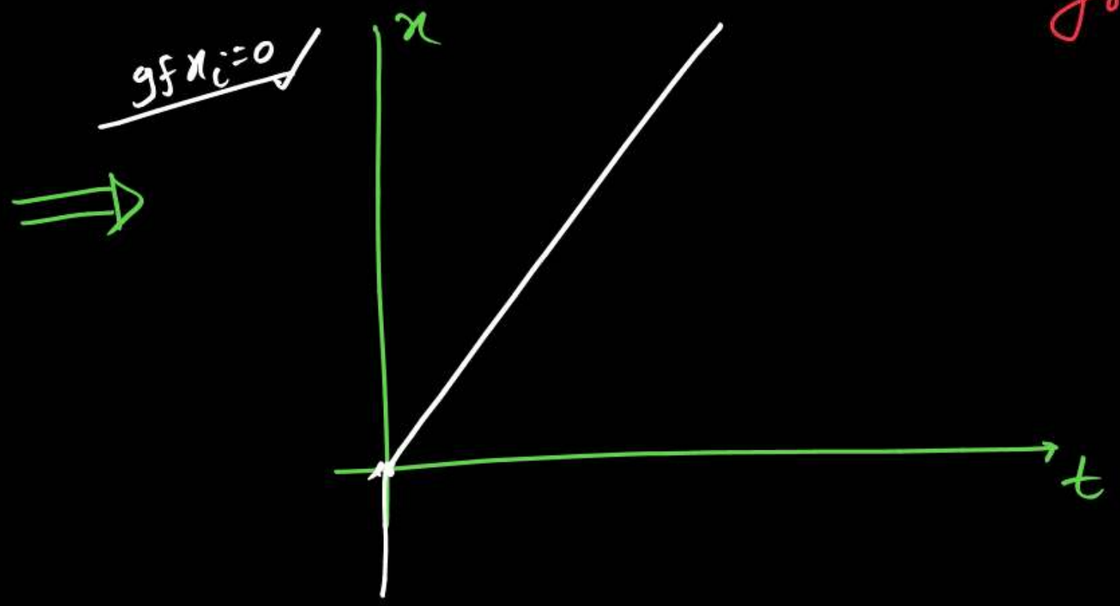
MR* Box

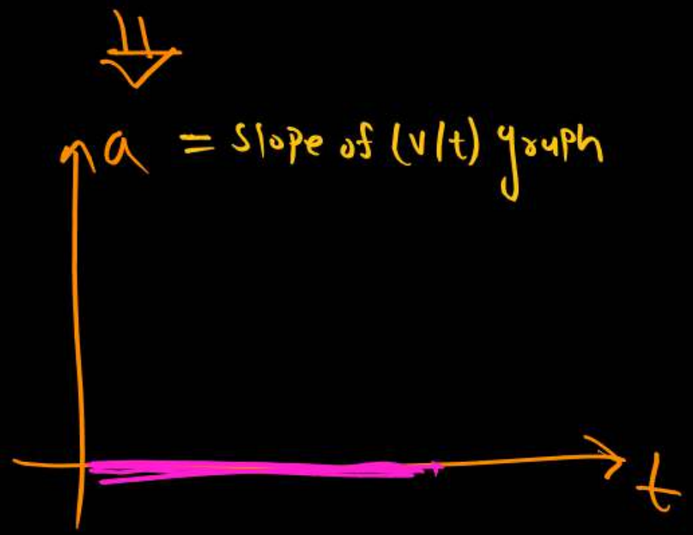
* In graph Conversion Nature of motion remains Same

* graph Conversion me given graph ya Jo convert Karna hai graph uske (Slope) pe focus Karo. **

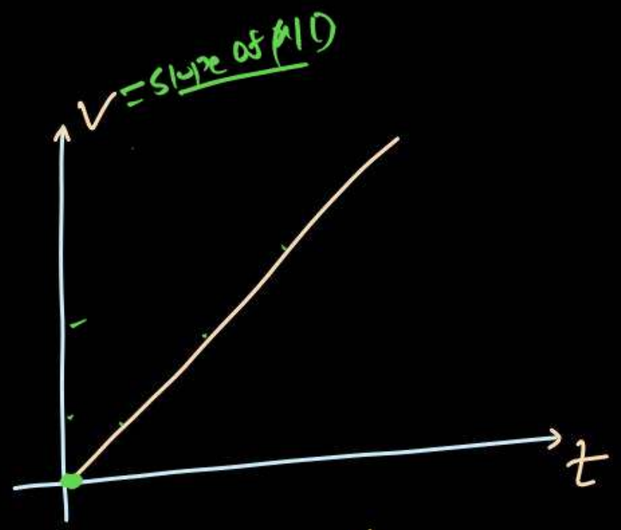
(Q) (56) Convert velocity time graph into Position - time and acc^{n} - time graph.



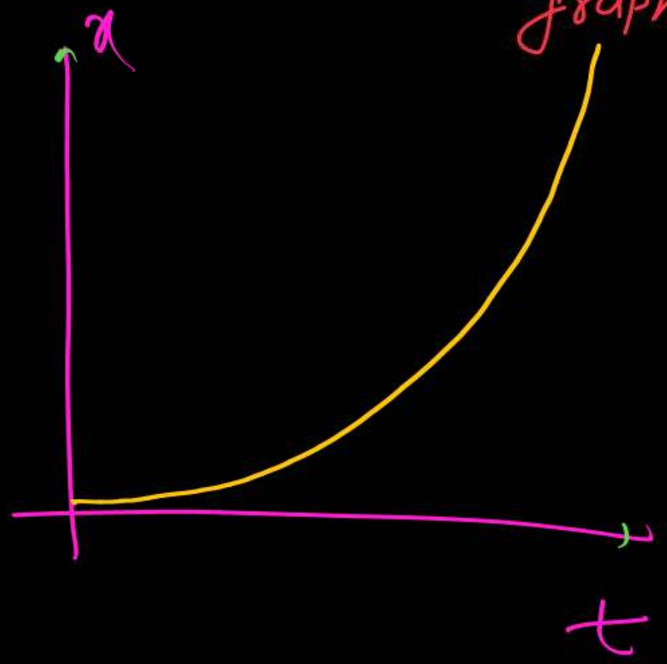


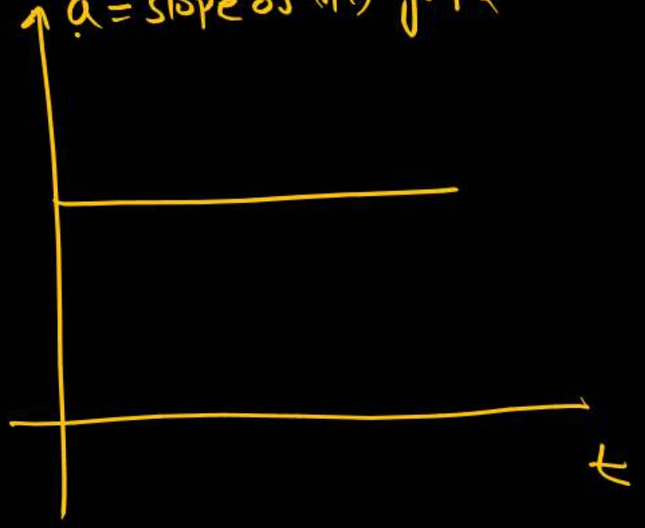


(Q) (57) Convert velocity time graph into Position - time and acc^{n} - time graph.

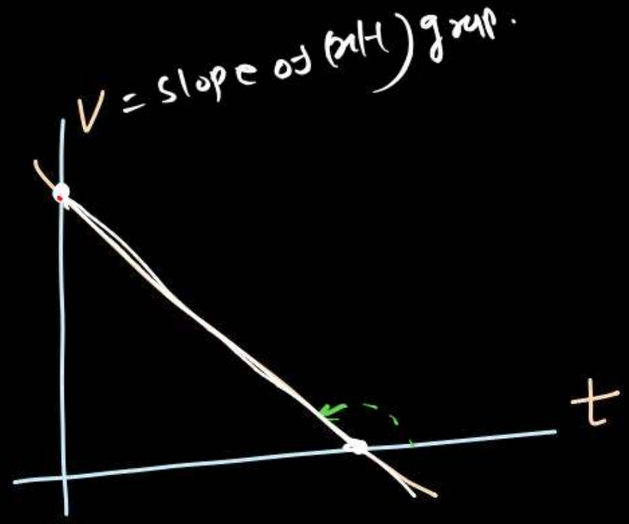




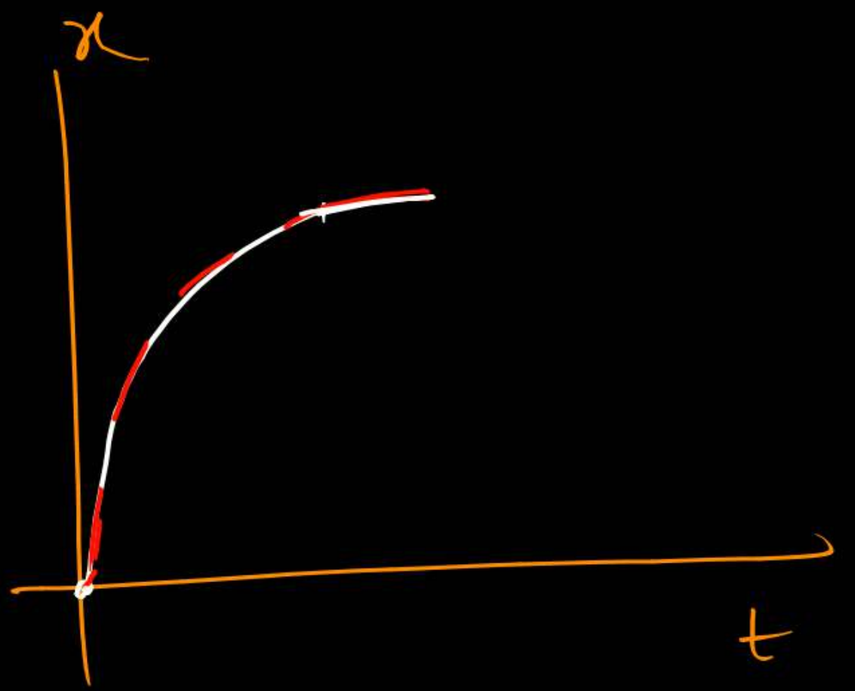


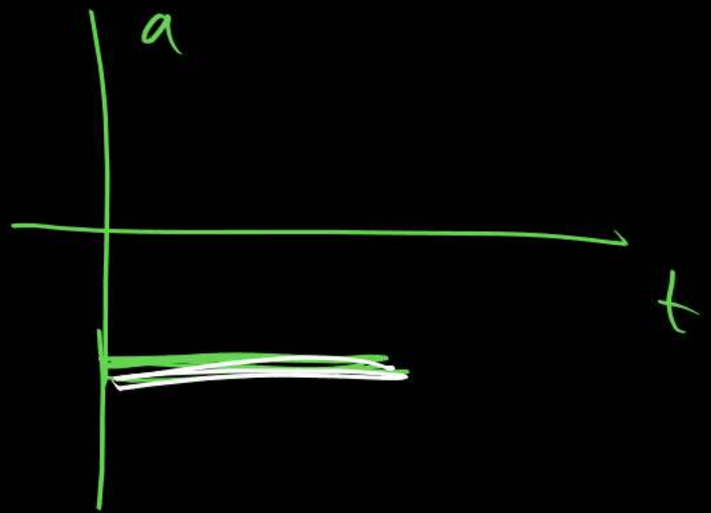


58 (Q) Convert velocity time graph into Position-time and acc^{n}-time graph.







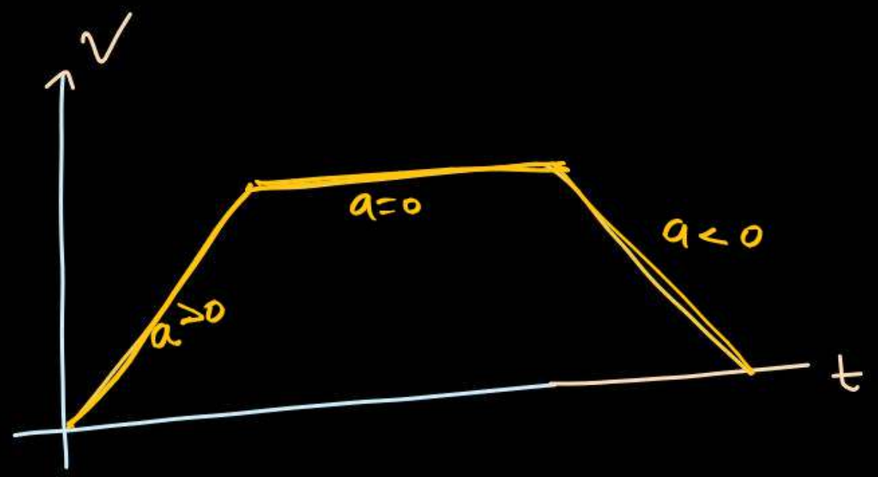


H/W

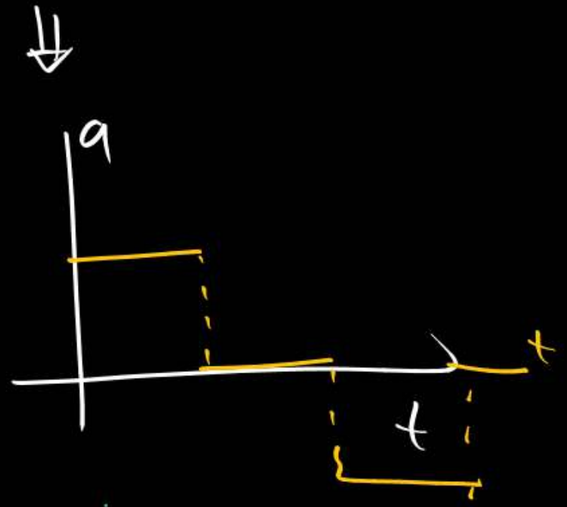
Solve in rough

copy in notes

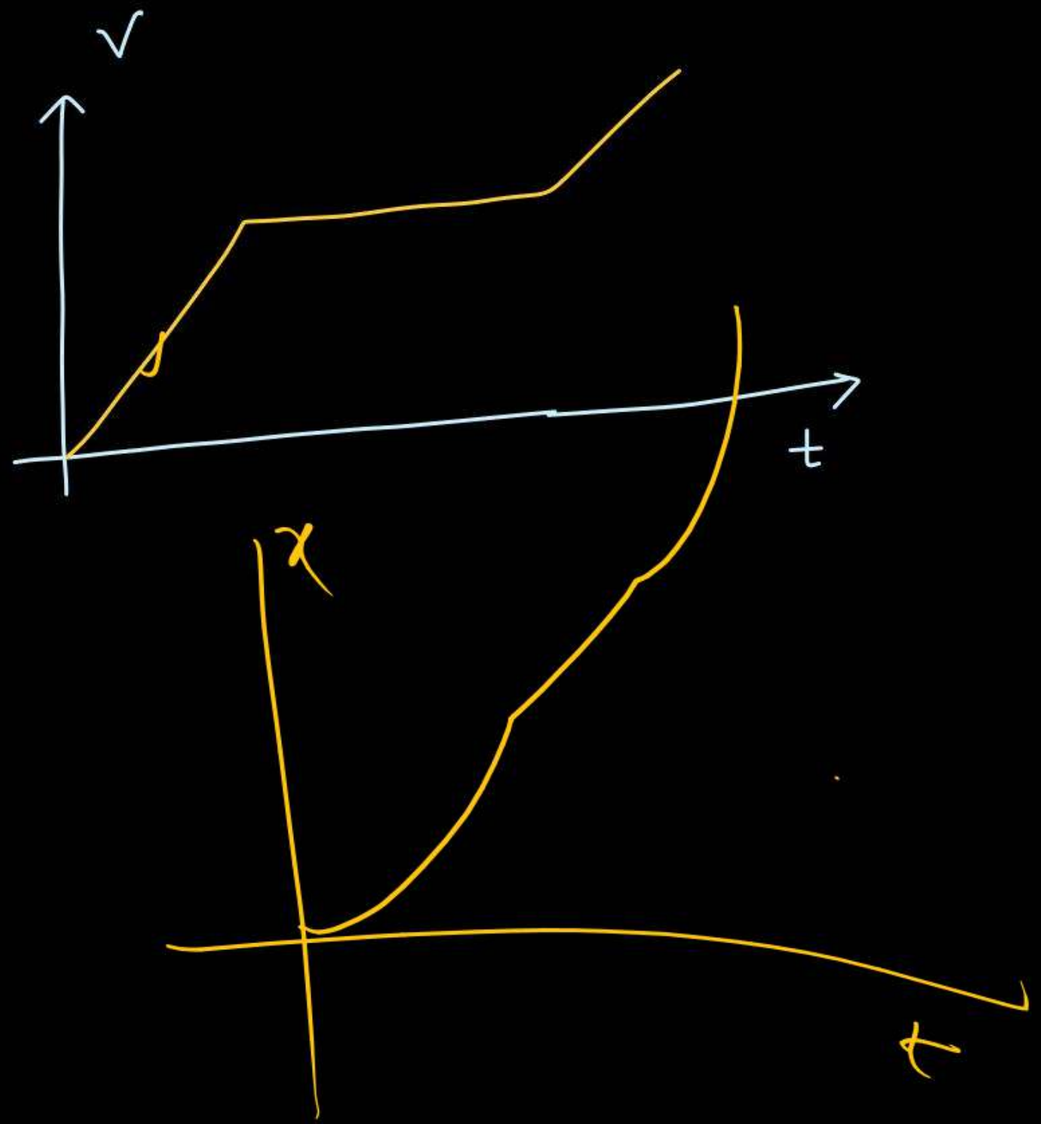
(Q) (59) Convert velocity time graph into Position-time and acc^{n}-time graph.





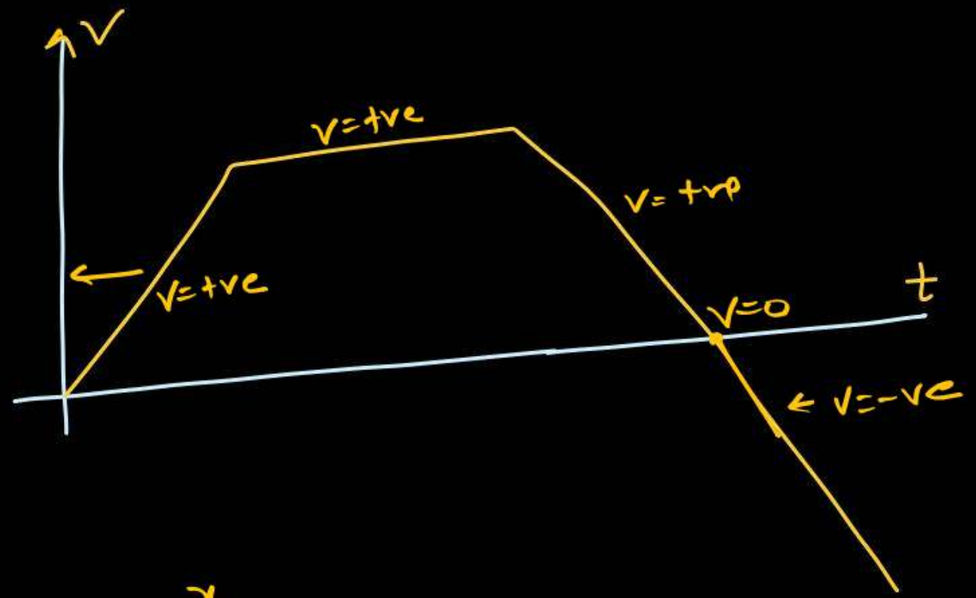


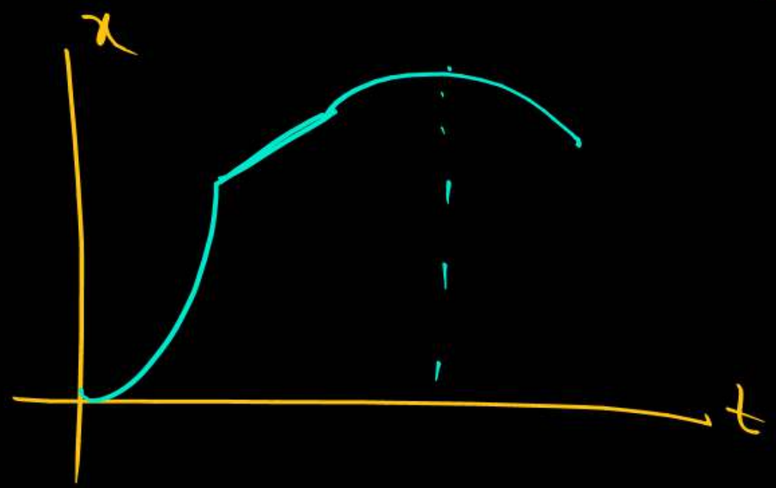
(Q) 60 Convert velocity time graph into Position-time and acc^{n}-time graph.



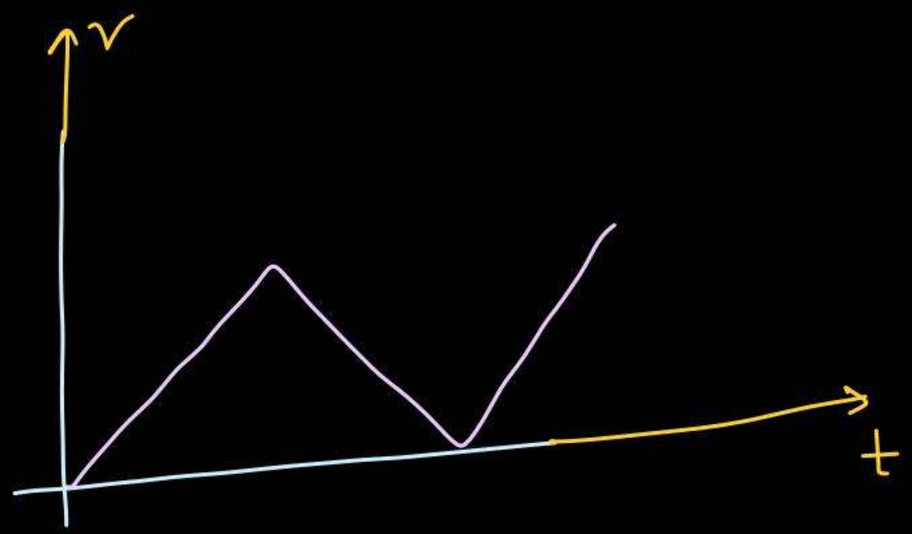
(Q) Convert velocity time graph into Position-time and acc^{n}-time graph.

61

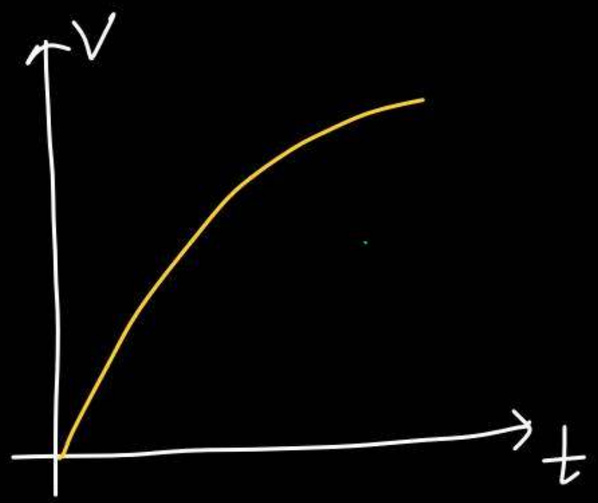


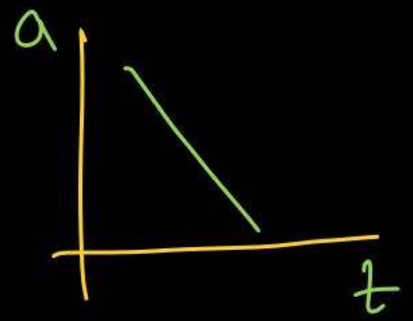


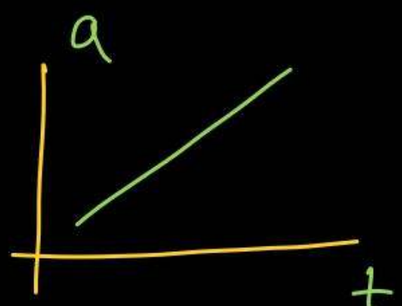
(Q) 62 Convert velocity time graph into Position-time and acc^{n}-time graph.

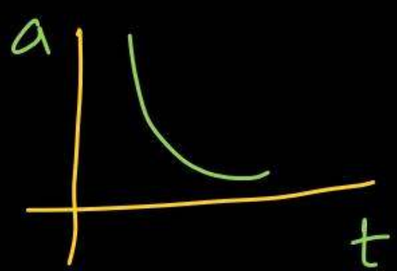


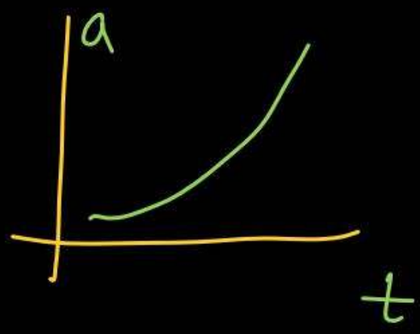
63 for given velocity-time graph which of the following acc-time graph is correct



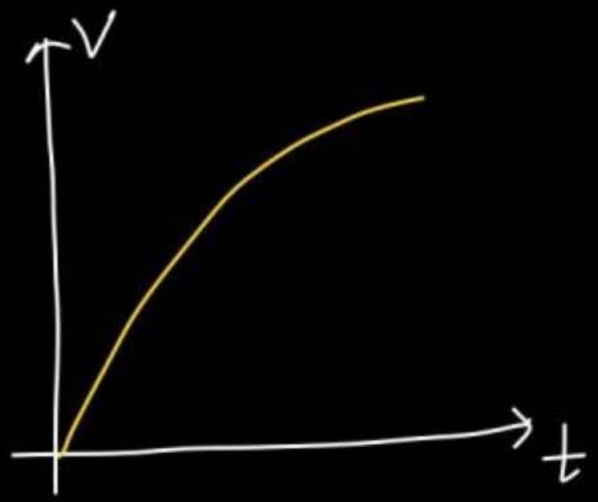


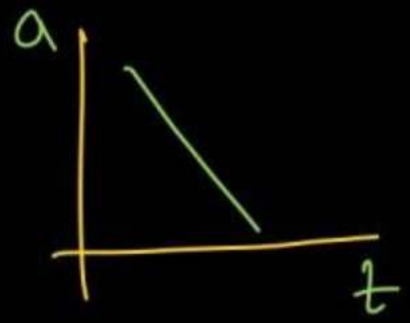


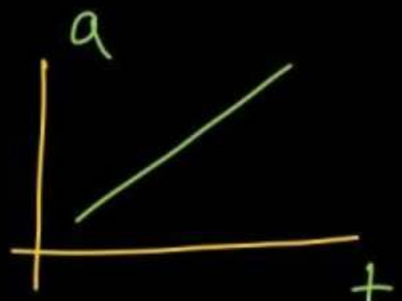


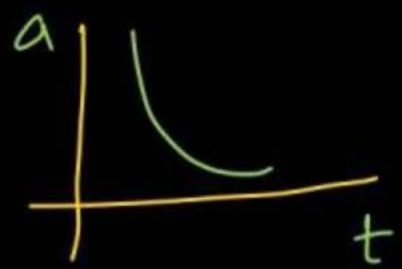


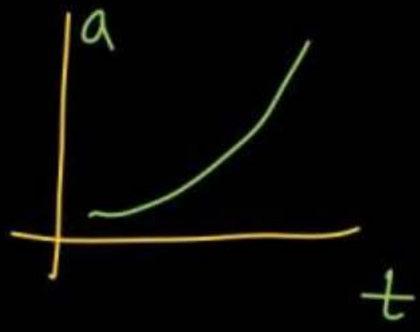
64 for given velocity-time graph which of the following acc-time graph is correct







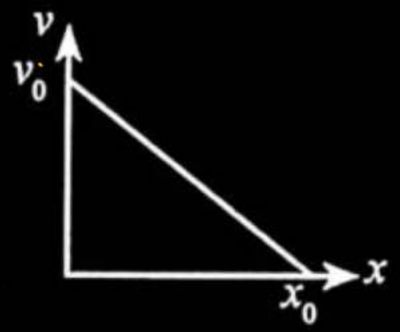




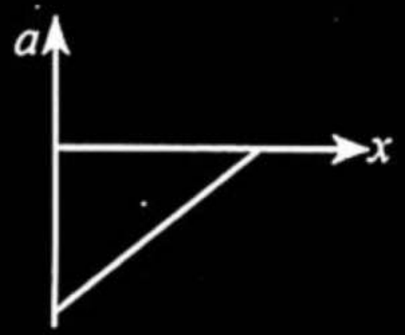
Question 65

The given graph shows the variation of velocity with displacement. Which one of the graph given below correctly represents the variation of acceleration with displacement?

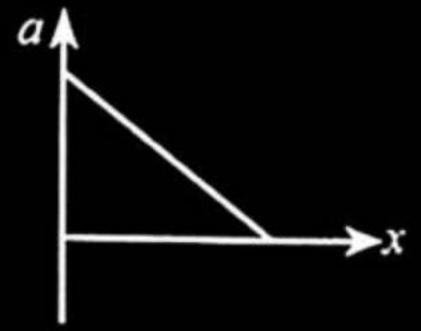
[JEE Adv. 2015]



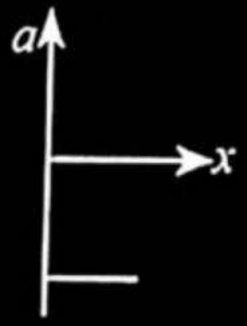
1



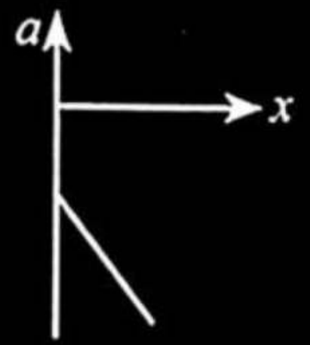
2



3

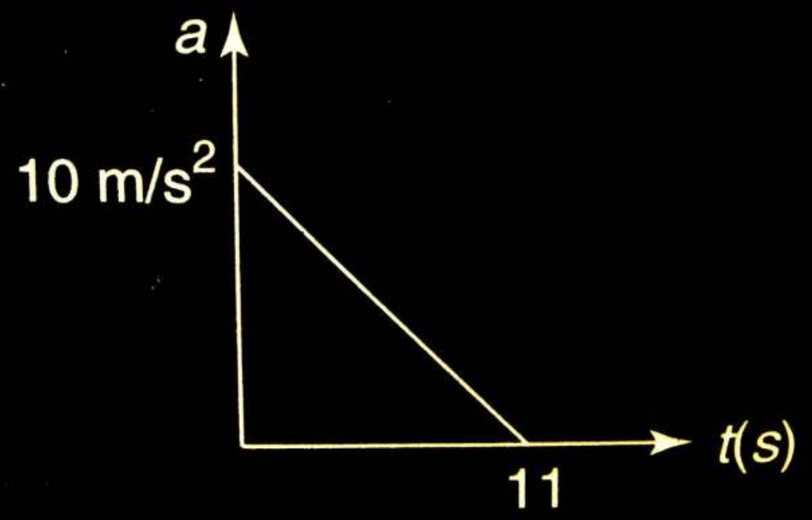


4



Question 66

A particle starts from rest. Its acceleration (a) versus time (t) is as shown in the figure. The maximum speed of the particle will be **[IIT-JEE (Screening) 2004]**

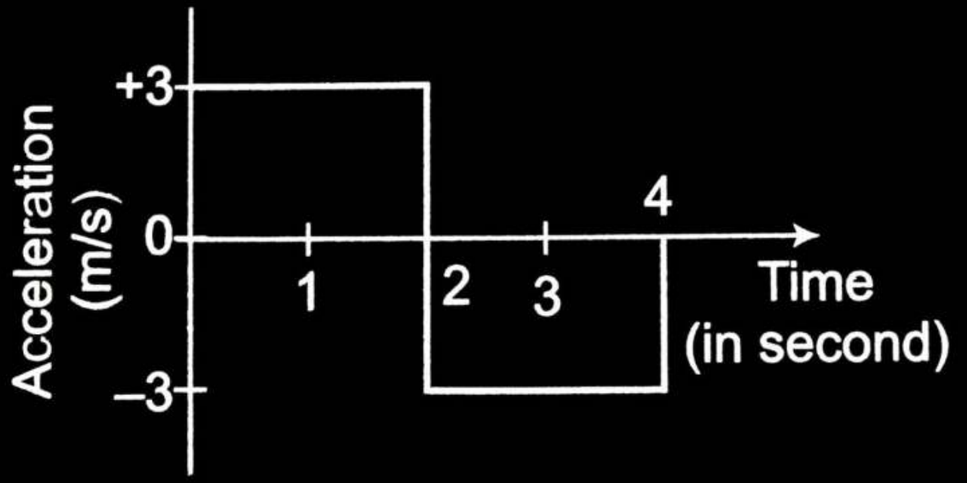


1. 110 m/s
2. 55 m/s
3. 550 m/s
4. 660 m/s

Question

67

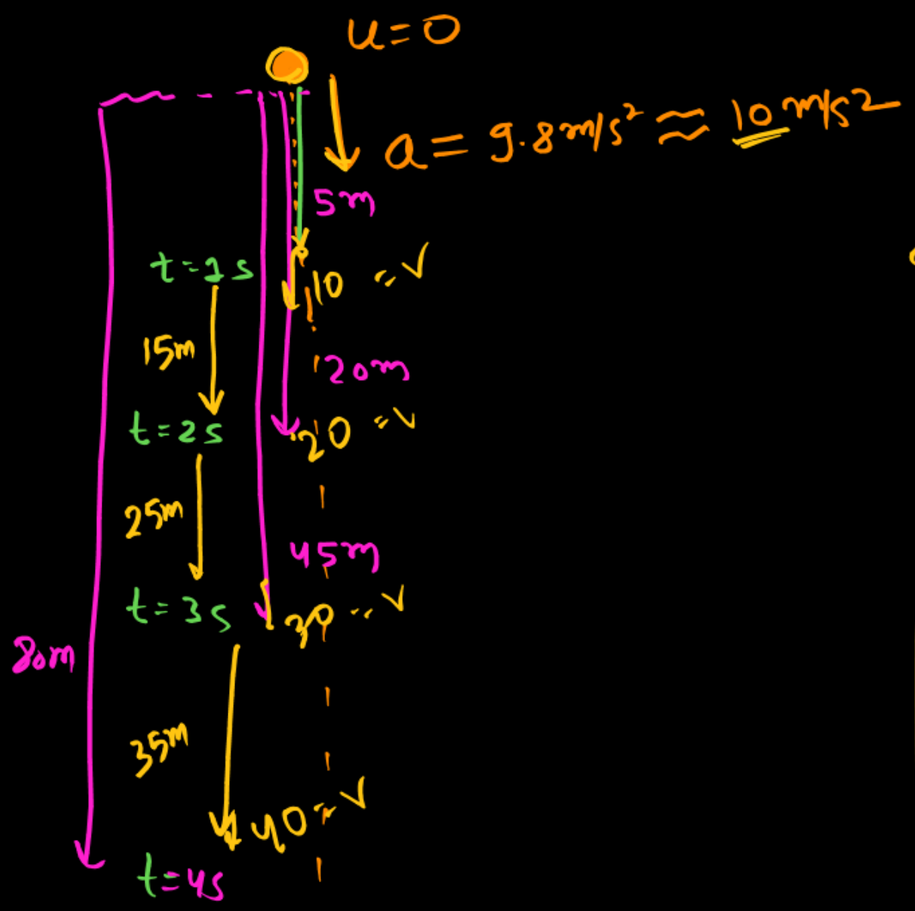
A particle starts from rest at $t = 0$ and moves in a straight line with an acceleration as shown below. The velocity of the particle at $t = 3s$ is

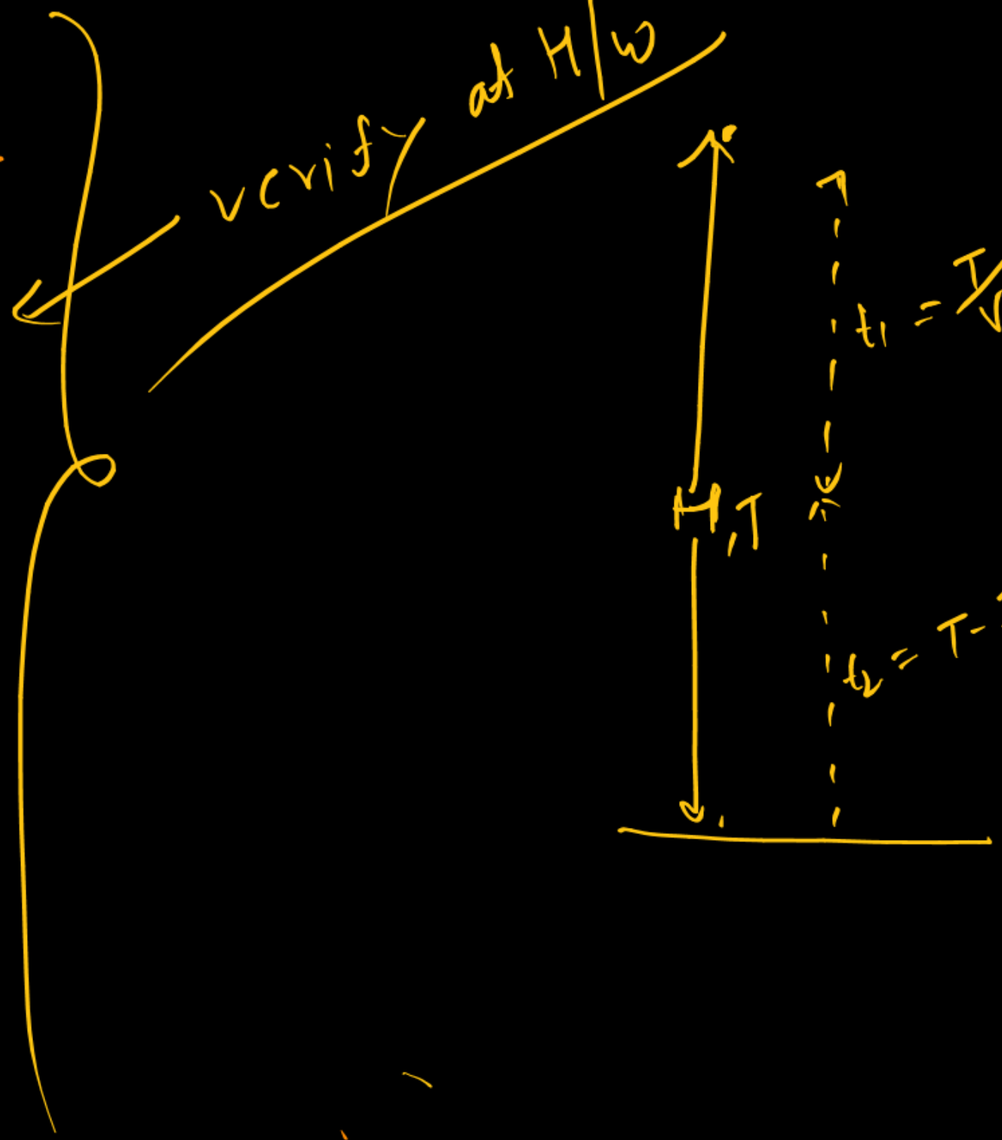


1. 2 m/s
2. 3 m/s
3. 4 m/s
4. 6 m/s

@ Motion starts from rest & const^n acc^n = 10 m/s^2 then find disp^m in 1s, 2s, 3s, 4s, 5s...

Motion under gravity





Thank

You